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IN-VITRO ANTIFUNGAL SUSCEPTIBILITY REVEALS OCCURRENCE OF AZOLE AND ALLYLAMINE RESISTANCE AMONG CLINICAL ISOLATES OF *CANDIDA ALBICANS* AND *CANDIDA NON ALBICANS* FROM CENTRAL INDIA

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ABSTRACT: Background: Drug resistance among *Candida* species constitutes the most significant problem in the treatment of Candidiasis. Systematic studies on antifungal drug susceptibility which may be useful in deciding clinical strategies are not routinely done in India and other developing countries.

Objective: Aim of this study was testing sensitivity of clinical isolates of *Candida albicans* and non *albicans* to frequently prescribe antifungal drugs, fluconazole, ketoconazole, Itraconazole, Amphotericin B and Terbinafine.

Material and methods: 25 strains of *C. albicans* and non *albicans* were tested *in vitro* for susceptibility to five antifungal agents, by using standard broth macro dilution method (CLSI M 27-A). **Results:** The present study revealed the percentage and extent of emerging drug resistance and cross resistance among Central Indian clinical isolates of *C. albicans* and non *albicans* against azoles and allylamine. In the present study, the total percentage of resistance was found to be 84% (21/25). The drug for which maximum resistance were found was Ketoconazole (64%) followed by Itraconazole (44%) Terbinafine (24%) and lastly Fluconazole (20%). The total percentage of cross resistance was 62% (13/21) and the maximum seen in *C. albicans* followed by *C. glabrata*, *C. krusei* and *C. guilliermondii*. No resistance was found for polyene drug Amphotericin B. **Conclusion:** This short study from India has exhibited the increasing frequency of resistance and Cross resistance of *Candida* species against azoles and allylamine. We suggest a comprehensive study to determine the extent and degree of antifungal drug resistance among *Candida* species in India.

INTRODUCTION: The genus *Candida* includes several species implicated in human pathology. *Candida albicans* is by far the most common species causing infections in humans. The emergence of non-*albicans Candida* species as a significant pathogens has however been well

Candida spp. have been shown to cause a similar spectrum of disease ranging from oral thrush to invasive disease, yet differences in disease severity and susceptibility to different antifungal agents have been reported ^{2, 3} Immunocompromised population is increasing throughout the world as